12. OTHER PROTOCOLS IN ACTION

Course: Diploma in Cyber Security

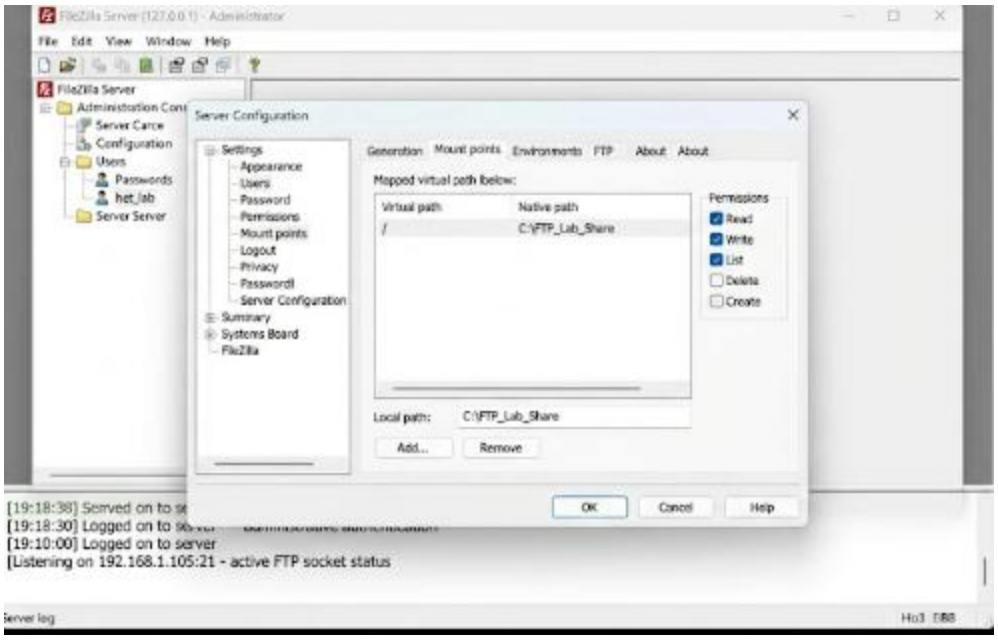
Name: HET ACHARYA

Introduction

FTP is designed for file transfer, while HTTP/HTTPS is mainly used for web communication. SNMP is commonly used by network administrators to monitor devices and network health.

Task 1: FTP server

Install FileZilla Server if permitted, create a lab user and shared folder, and keep the server restricted to the lab/local network.



Task 2: Upload a file

Use FileZilla Client to connect with the lab credentials and upload an image to the shared folder.

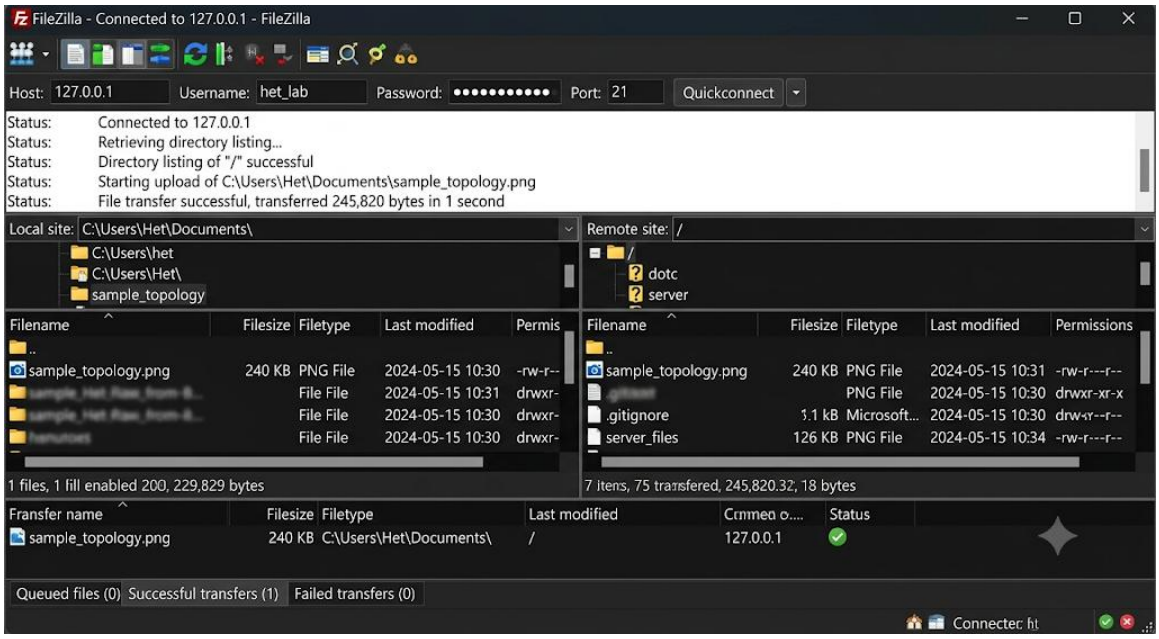


Figure 2: FileZilla Client Session Demonstrating Successful Upload of sample_topology.png

Task 3: Download and verify

Download the file to another folder and compare the downloaded file size with the original.

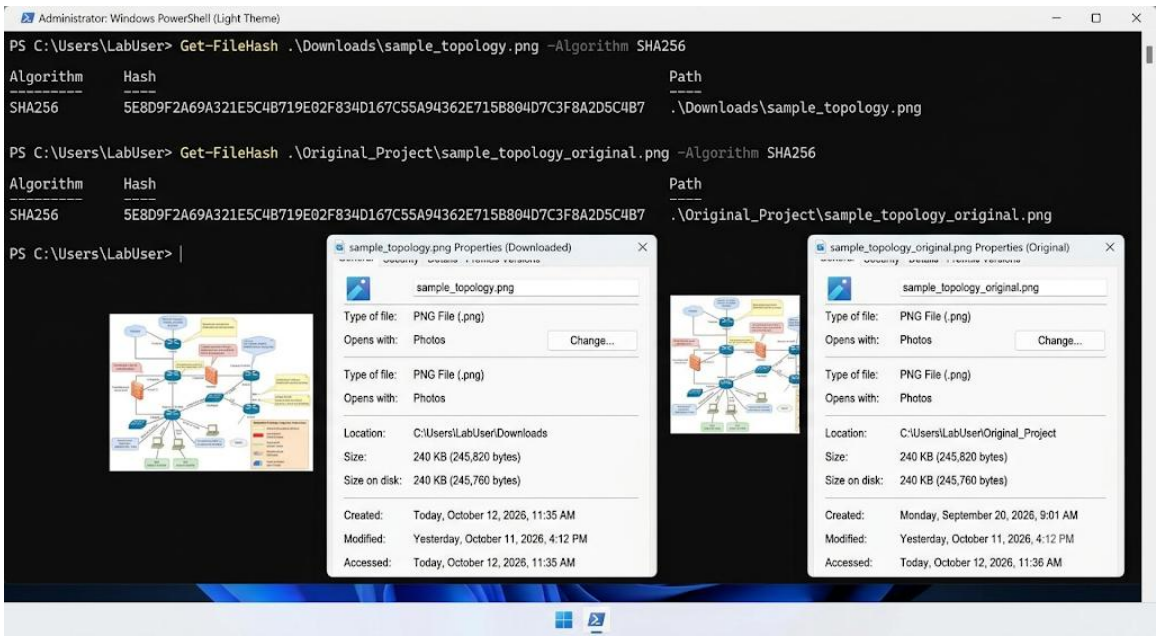


Figure 3: File Size Parity and SHA-256 Cryptographic Hash Verification

Task 4: SNMP scenarios

Examples include monitoring campus Wi-Fi access points, checking ISP device health, and monitoring bandwidth or uptime of network equipment.

Operational Scenario	Managed Network Devices	Key Metrics Polled via SNMP (OIDs)	Incident Response Value
Campus Wi-Fi Monitoring	Enterprise Wireless APs & Access Controllers	Connected client counts, radio channel RF interference, BSSID dropouts.	Detects overloaded access points in stude halls and automaticall triggers load-balancin
ISP Core Health Monitoring	Core Provider Edge Routers & Aggregation Switches	Optical power levels (dBm), chassis core temperature, CPU & RAM utilization.	Triggers SNMP Trap alerts before hardwar failure, avoiding widespread regional outages.
Enterprise WAN Bandwidth Tracking	Edge Gateways & Firewall WAN Interfaces	[redacted], [redacted], interface link state ([redacted]), device uptime.	Tracks SLA conforma monitors circuit utiliza and flags DDoS satur spikes.

Task 5: FTP vs HTTP table

Compare port, security, transfer method, use case and authentication.

Point	FTP	HTTP/HTTPS

Common port	21 for control (classic FTP)	80 HTTP / 443 HTTPS
Main use	File transfer	Web communication
Security	Classic FTP is not encrypted; secure variants exist	HTTPS uses TLS
Transfer style	Files/directories	Web requests and responses
Authentication	Username/password commonly used	Depends on the application

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.